

Chapter 18 — Fundamentals of Laparoscopic and Robotic Urologic Surgery

- **Osler: "Diseases that harm require treatments that harm less."** Over the past 30 years, minimally invasive urology has become **predominant**.
- Multi-institutional comparisons with open surgery show **equivalent efficacy and acceptable efficiency**. The advantages are **less pain, better cosmesis, faster recovery, shorter stay** and, as techniques mature, often **lower cost**.
- The objectives of even complex open surgery on the adrenal, kidney, ureter, bladder, prostate and lymph nodes can now be met **with less injury**. Open surgery's role is steadily shrinking.

PREOPERATIVE PREPARATION

Patient Selection and Contraindications

- Start with a **history focused on previous surgery** and an **examination mapping every abdominal scar**. Laboratory tests, electrocardiogram and chest radiograph follow the same criteria as any major operation under general anaesthesia.
- **Severe chronic obstructive pulmonary disease (COPD) → arterial blood gases and pulmonary function tests**, because the CO₂ pneumoperitoneum loads the lungs. **Arrhythmias must be treated beforehand**: hypercarbia and acidosis destabilise the myocardium.

Contraindications	Conditions demanding caution and specific consent
• Uncorrectable coagulopathy • Intestinal obstruction (unless it is to be treated) • Significant abdominal wall infection • Massive haemoperitoneum or haemoretroperitoneum • Generalised peritonitis • Suspected malignant ascites	Morbid obesity · extensive previous surgery · pelvic fibrosis · organomegaly · benign ascites · pregnancy · hernia · iliac or aortic aneurysm · surgeon inexperience

DIFFICULT PATIENT GROUPS AND HOW TO MANAGE THEM

Condition	Problem	Management
Morbid obesity	Instruments too short; limited trocar range; higher pressures needed ; poor orientation	Laparoscopic renal and adrenal surgery is still better than open (blood loss, oral intake, analgesia, stay, convalescence), including partial nephrectomy and nephroureterectomy. Prostatectomy and cystectomy carry more complications with body mass index (BMI) >30 → schedule later in the surgeon's experience
Extensive previous surgery	Adhesions	Palmer point Veress entry (left subcostal, midclavicular line), open Hasson access , or a retroperitoneal start, entering the peritoneum from behind

Condition	Problem	Management
Pelvic fibrosis	Previous peritonitis, pelvic surgery, endometriosis; hip prosthesis cement (polymethyl methacrylate) leak → dense fibrosis during pelvic node dissection	Anticipate a severe technical challenge
Organomegaly	Liver or spleen at risk	Veress far from the organ, or Hasson
Benign ascites	Bowel close to the peritoneum; leakage	Watertight closure , firm dressing
Pregnancy	Uterus; caval compression; fetal acidosis	Access cephalad , away from the fundus (often the Palmer point); pressure 8–12 mm Hg (≥ 15 mm Hg risks hypotension); second trimester preferred ; feasibility falls after 20 weeks
Hernia	Diaphragmatic hernia → CO₂ into the mediastinum (rarely tamponade or respiratory compromise)	Never insufflate through an umbilical or abdominal-wall hernia site
Iliac or aortic aneurysm	Puncture risk	Vascular review; left upper quadrant Veress or Hasson; secondary trocars under vision
Surgeon experience	Learning curve	Robotic surgeons with more than 20 cases have better outcomes; laparoscopic experience improves robotic partial nephrectomy; use proctoring

Bowel Preparation

- **No mechanical bowel preparation** for retroperitoneoscopy, extraperitoneoscopy, or transperitoneal surgery not using bowel. Propensity-matched data show **no benefit** for laparoscopic nephrectomy or prostatectomy, and emerging data show none even for **cystectomy with conduit or neobladder**.

Enhanced Recovery After Surgery Protocols

- **Enhanced recovery after surgery (ERAS)**, Europe, **2001**: a multimodal, multidisciplinary programme.
- **Elements: no solids for 6 hours and carbohydrate drink 2 hours before surgery** (not overnight fasting); **balanced, not liberal, fluids**; early removal of drains and tubes; early mobilisation; **eating and drinking on the day of surgery**. Later additions: counselling, thromboprophylaxis, ileus prevention, normothermia.
- **Results**: stay shortened by about **30%**, complications halved, fewer readmissions, lower cost. Benefit in cystectomy is seen with **open or minimally invasive** approaches. **Surgery-anaesthesia collaboration** is the key.

Preparation of Blood Products

- **Group and screen suffices** for low-bleeding-risk procedures: radical nephrectomy and nephroureterectomy (transfusion **3–12%**), prostatectomy (**2.5%** at experienced centres).
- **Crossmatched blood** for extensive procedures (partial nephrectomy, cystectomy, caval thrombectomy), **especially early in experience**. With experience, group and screen may do for partial nephrectomy (transfusion **6–7%**).

KEY POINTS: PREOPERATIVE PREPARATION

- **Patient selection** starts with surgical history and a **scar map**.

- **Contraindications:** uncorrectable coagulopathy, untreated intestinal obstruction, abdominal wall infection, massive haemoperitoneum or haemoretroperitoneum, generalised peritonitis, suspected malignant ascites.

Concept — Every relative contraindication is a problem of **safe entry or tolerance of the pneumoperitoneum**. The answers recur throughout the chapter: enter elsewhere (Palmer point, retroperitoneum), enter openly (Hasson), or insufflate less.

OPERATING ROOM

Setup of the Operating Room

- Positions of equipment and staff are **defined for each case**. All equipment must be **checked before starting** (Box 18.1), with an **open laparotomy tray ready**.

BOX 18.1 INSTRUMENTATION CHECKLIST FOR MAKING A SKIN INCISION FOR OBTAINING THE PNEUMOPERITONEUM

- Irrigation–aspiration working · electrosurgical unit working · **CO₂ tank full, with a spare in the room** · camera **white-balanced**, light source working · insufflator checked for flow and for its **response to tubing kinking** · Veress needle checked for flow and tip retraction.

Patient Positioning

- **Supine:** arms tucked or on sleds. **Trendelenburg or lateral:** tape and belts across chest and thighs.
- **Lateral:** pad all bony prominences and the strap contact points. **Lower leg flexed about 45°**, upper leg **straight** on pillows (no sciatic stretch). Pad the **knee and ankle** of the lower leg. **Axillary roll three to four fingerbreadths below the axilla** (brachial plexus). Use active warming for long cases.
- **Flank interface pressure** is higher in men, with **BMI ≥25**, with the **kidney rest** (the most harmful; avoid beyond 20–30 minutes), and with **full rather than half table flexion**. → **Men with BMI ≥25, kidney rest up and full flexion are at highest risk of rhabdomyolysis**. The **plain table mattress beat egg-crate or gel**, and egg-crate was at least as good as the costlier gel.
- **Pelvic surgery:** Trendelenburg on **split-leg positioners or Allen stirrups**. **Never shoulder braces** (brachial plexus injury).

Cord Management

- Secure cords to the drape **at two points** without compressing them. Keep the **camera cable separate** from diathermy and suction. Place the boom or robotic cart at the **foot of the table** without cable tension. On the robot, **wrap redundant camera cord once round the camera shaft**. Leave a **U-shaped loop** in suction tubing when racked.

Prophylaxis and Other Preparations

- **Pneumatic compression stockings;** optional **5,000 units subcutaneous heparin** preoperatively; **both in high-risk patients** (e.g., morbid obesity), continued postoperatively.
- **Foley catheter:** monitors urine output, decompresses the bladder (space, fewer pelvic injuries). Secure it with a stabiliser and put the bag at the **head of the table**.
- **Nasogastric or orogastric tube** for upper-abdominal working space, especially on the **left**, where a full stomach hampers spleen mobilisation and access to the kidney.

STRATEGIC PLACEMENT OF OPERATIVE TEAM AND EQUIPMENT

Standard Laparoscopic Carts and Integrated Endoscopy Systems

- **Traditional cart:** insufflator, light source, camera controls, recorder. **Insufflator at the surgeon's eye level** for continuous pressure monitoring.
- **Integrated systems:** flat panels on **ceiling booms** placed directly before the surgeon, reducing eye and body strain; the tower can sit anywhere.

Robotic Systems

- **da Vinci (Intuitive Surgical)**, three components: **patient-side cart** (arms and instruments), **surgeon's console**, and **vision cart** (monitor, energy generator, light source, camera electronics).

DA VINCI PLATFORMS COMPARED

	Si	Xi	SP (single port)
Camera	12-mm port	8-mm high-definition (HD) camera in any port (multiquadrant work)	Articulating camera
Cart	Fixed approach (pelvis: between the legs)	Boom-mounted; any side of the table; laser targeting; pelvic cases without stirrups	Single arm: no external collisions
Arms and instruments	Bulkier	Finer arms → flexible port placement, fewer collisions	Three instruments with elbow and wrist; 360° access to all quadrants

Placement of the Operative Team for Laparoscopic Procedures

TEAM PLACEMENT BY APPROACH

Approach	Laparoscopic	Robotic
Transperitoneal, upper abdomen (modified lateral decubitus)	Surgeon and assistant facing the abdomen (left nephrectomy → patient's right side); scrub nurse opposite, near the foot. A 6 × 4 ft × 6 in platform keeps the elbows at the sides for suturing	Assistant facing the abdomen; cart from the back , angled towards the head so the camera points at the target (fewer clashes); scrub nurse on the assistant's side
Retroperitoneal, upper abdomen (true 90° lateral, table flexed at the hip to open the 12th rib–iliac crest gap)	Surgeon and camera assistant facing the patient's back ; scrub nurse in front	Robot over the patient's head → anaesthetist moved to the side, in front
Pelvic (supine, split legs or stirrups, slight hip flexion, arms tucked, chest strap, 30° Trendelenburg)	Surgeon on the preferred side, assistant opposite	Si: robot between the legs ; scrub nurse on the assistant's side. Xi: robot from either side, no stirrups

Figure legends, Figs 18.1, 18.2, 18.6, 18.8, 18.9 — Split-leg positioners and Allen stirrups; cables fixed at two points with a U-loop in the suction tubing; operating-room layouts for each approach, as tabulated.

PERFORMING THE PROCEDURE

Before the Initial Incision

- Complete **Box 18.1 just before insufflation**. For the robot: console, vision cart and patient cart **plugged into different circuits**; cables connected; **self-test and homing** complete; 3D endoscope **calibrated, black/white-balanced, target-aligned**; patient cart draped.

Achieving Transperitoneal Access and Establishing the Pneumoperitoneum

Closed Techniques

- **Veress needle**: 70- or 120-mm, 14-gauge, 2-mm outer diameter. **Pre-checks**: blunt tip springs back; saline passes; at 2 L/min the pressure stays ≤ 2 mm Hg; no leak at the hub when the tip is occluded.
- Grasp it at midshaft and pass it perpendicularly. **Two points of resistance: fascia, then peritoneum**.

VERESS NEEDLE ENTRY SITES

Site	Advantage	Danger and adjustment
Superior umbilical border (head down 10–20°)	Thinnest wall; best cosmesis	Great vessels (left common iliac, aorta, cava). Non-obese : angle towards the pelvis (the umbilicus overlies the bifurcation). Obese : perpendicular (the umbilicus migrates caudally)
Lower quadrants (supine)	Less vascular risk	Colon
Lateral decubitus	—	Two fingerbreadths medial and two superior to the anterior superior iliac spine
Palmer point (left subcostal, midclavicular line; or the right equivalent)	Preferred with suspected adhesions	Too deep → liver (either side) or spleen
Scarred abdomen	—	Use an unscarred quadrant ; if none → open access

- In a large series of blind Veress placement (including previous surgery and obesity), there were **no major vascular or hollow-organ injuries**, only minor liver lacerations managed conservatively.

Confirming placement: (1) aspiration-irrigation-aspiration: 10-mL syringe with 5 mL saline; aspirate (blood or bowel particles?) → inject freely → re-aspirate (nothing returns) → inject 2–3 mL more to clear omentum from the tip → **(2) drop test**: saline in the open hub falls freely → **(3) advancement test**: advance **1 cm without resistance** (resistance = still preperitoneal) → **only then insufflate**, at up to **9 L/min to 15 mm Hg**. Free flow is confirmed by an initial pressure **<8–10 mm Hg**, with flow stopping at 15 mm Hg.

- Towel clips or Allis clamps can steady the fascia, but **do not lift the fascia**: it widens the fascia-peritoneum gap without enlarging the abdominal space.

Open-Access Techniques

- **Hasson**: easier and, early in experience, safer. It needs a larger incision and **leaks gas more**. **Recommended when extensive adhesions are expected**; it is as efficient as closed access and possibly safer after previous surgery.

Hasson sequence: **2-cm semicircular infra-umbilical incision** → open fascia and peritoneum **separately**, transversely, finger-sized → confirm entry by sight and touch → **two 0 silk fascial stay sutures** → insert the cannula with the blunt tip → seat the funnel adapter and wrap the sutures round its struts → remove the obturator → **insufflate at maximum flow**.

- **Hand-port access**: safest for the inexperienced. Open a **6.5–7.5-cm** incision, place the device, insufflate through it, and guide secondary trocars with the intra-abdominal hand (or place them under vision).

Retroperitoneal Access and the Retroperitoneal Space

Open (Hasson) retroperitoneal access (the commonest, most precise): **2–2.5-cm transverse incision in the midaxillary line just below the tip of the 12th rib** → S-retractors → incise the posterior lumbodorsal fascia, split muscle → open the **anterior thoracolumbar fascia** snug to the index finger (limits leak) → **finger confirms the psoas behind and the Gerota-covered lower pole in front** → finger-develop a space → **balloon inflated twice, cephalad then caudad**, anterior to the psoas and **posterior to Gerota's fascia** → extra caudal dilatation for ureteric work; extra cephalic subdiaphragmatic dilatation for adrenalectomy.

BALLOON DILATORS COMPARED

	Commercial preperitoneal balloon dissector	Self-styled (Gaur) dilator
Construction	Transparent silicone balloon on a stiff, hollow, clear shaft; room air by bulb; maximum 800 mL (40 pumps) . Oblong shape for the retroperitoneum, round for the pelvis	Gaur (1992) : size 7 glove on an 8-Fr red rubber catheter, inflated to 110 mm Hg. Variants: glove finger(s), condom, drape cot on a 16–18-Fr catheter, tied with 0 silk, filled with saline , back-loaded into a lubricated 30-Fr Amplatz sheath
Advantages	Directable; laparoscope inside the shaft confirms position and dilatation	Cheap
Drawbacks	Cost	No stiff shaft, no endoscopic view; latex ruptures into many fragments (Silastic leaves one, easier to retrieve)

- **Complications** come from misplacement or rupture: **intramuscular dilatation** → **hernia**, or inadvertent peritoneal tears.

Extraperitoneal Access and the Extraperitoneal Space

Pelvic extraperitoneal access: **1.5–2-cm infra-umbilical incision** → anterior rectus sheath opened vertically 1.5 cm → recti separated → finger tunnels caudally between the rectus and posterior sheath to the pubis → **puncture the transversalis fascia** → digital prevesical space → balloon in the midline, then each side → prevesical fat displaced, peritoneum reflected cephalad.

- **Caveat**: early retroperitoneoscopy with a standard Hasson cannula caused **heavy subcutaneous emphysema and hypercarbia**. This was solved by the **blunt balloon port**: an internal balloon plus an external foam cuff give an airtight seal.

Access Technology: Trocars, Hand Ports, and Single-Port Access

Trocars

- A trocar is a **cannula (port) plus obturator**; 3–20 mm diameter, 5–20 cm long. **One-way valves** allow instrument exchange; **multiseal valves** take 5 mm and larger without leak. Remove the seal to extract tissue.

Bladed (cutting) trocars	Blunt (dilating, non-bladed) trocars
Cut through tissue; needed a spring safety shield. Now of historical interest only	Spread the muscle; no shield needed; some optical (clear tip, endoscope inside)

Bladed (cutting) trocars	Blunt (dilating, non-bladed) trocars
More epigastric injury and port-site hernia	About fivefold less inferior epigastric injury; meta-analysis of randomised trials: trocar-site bleeding 3% vs 9%, overall complications 3% vs 10%
Fascial closure of all 10–12-mm sites (5-mm sites in children)	Fascial closure usually only for midline ports >10 mm , or any stretched site

- **Robot:** Xi camera fits any **8-mm** port; **Si camera needs 12 mm**. Instruments use proprietary **reusable 8-mm metal trocars with disposable valves**.
- **Reducers** are needed in flap-valve ports; **multiseal and AirSeal** valves take **5–12-mm** instruments without one.
- **Retention** (threaded sleeves, expandable arms, balloons) prevents dislodgement, leak and emphysema. The **Blunt Tip balloon trocar sandwiches the wall between an internal balloon and an external foam ring**, which is especially useful retroperitoneally.
- **AirSeal (CONMED): Intelligent Flow insufflator + valve-free port + tri-lumen filtered tubing**. It holds a **stable pneumoperitoneum despite suction or leak**, so routine work is at **8–12 mm Hg**. Smudge-free scope insertion, intact specimen extraction, continuous **smoke evacuation**.

Hand-Assist Devices

- **GelPort** and **Endopath Dextrus** maintain pneumoperitoneum with the hand inside. The cost falls on the surgeon: **more pain and numbness of hand, wrist and forearm**, and more fatigue, than standard laparoscopy.

Placement of Initial Trocar

- After Veress insufflation, hold the trocar in the dominant hand with the **middle finger along the shaft** and insert with a **twisting downward motion**, the other hand at the skin as a brake. **Perpendicular for mid/upper abdomen; 70° caudad for the pelvis**. The **hiss of CO₂** from the open side arm confirms entry. With an optical trocar, monitor entry with a 0° lens.
- Some raise the pressure to **20 mm Hg** for primary trocar insertion, to protect underlying structures. Data suggest it **may not increase volume or ease insertion**.
- With open access, the **Hasson cannula is the first trocar**.

Hand-Assist Placement

- As a secondary port: **insufflate first** (skin on stretch → smaller incision). Trace the device **“footprint”** and plan the other ports around it. **Incision length in cm = surgeon's glove size** (glove 7 → 7 cm).
- Waterproof the glove–gown seam with a drape or loban. Use a **brown glove** on the intra-abdominal hand (less glare).

Secondary Trocar Placement

- **Inspect the whole abdomen first** for access injury.
- Avoid **“crossing swords”** (tips meet) and **“rollover”**. Aim for **triangulation**: a **four-point diamond around the target**.
- **Suturing geometry**: needle drivers **<55° to the horizontal**, **25–45° between instruments**.
- **Point each port at the target**, so constant redirection does not widen the tract and cause emphysema.

- **Standard approach:** under vision with a **30° lens turned 180°** to view the wall. **Transilluminate** with lights dimmed to avoid wall vessels. Make a skin incision just wide enough; twisting insertion; suture-anchor non-retaining ports with No. 2 non-absorbable sutures.
- **Never use a metal trocar with a plastic retaining ring:** stray current cannot dissipate into the wall, so it injures viscera out of sight (capacitive coupling).
- **Hand-assist approach:** the intra-abdominal **finger guides each non-bladed trocar**, which is fast and safe.

Trocar Configuration and Robotic Considerations

- Detailed configurations are in Chapter 142 (upper tract) and Chapter 161 (pelvis). The rule: **minimise intra- and extracorporeal clashing**.
- **Robot:** camera **8 mm (Xi) or 12 mm (Si)**; arm ports 8 mm; a **12-mm port with an 8-mm reducer** for the robotic stapler. **Ports 8–10 cm apart**. Put the assistant's port **caudal to the arms, not between them**. Advance robotic ports until the **thick black line is below the abdominal wall**.

Laparoscopic Instrumentation

Instruments for Visualization

- **Four components:** laparoscope, light source, camera, monitor.
- **Laparoscopes:** 2.7–10 mm; **0° or 30°** (range 0–45°). The **30° lens** shows more and can look round structures. **Deflectable tip** (up to 100° in four directions; needs an adept assistant). **Variable-view prism** (0–120°, standard eyepiece).
- **Rod-lens with fibreoptic light** versus **chip-on-tip (CCD) digital scopes**, which give a better image and need no separate light cord.
- **Fogging:** pre-warm (warm saline or thermos), antifog solution or povidone-iodine, all-in-one defogger. Dry the eyepiece–camera junction (not an issue with digital scopes). **White-balance**.
- **3D: two-lens binocular** systems give depth perception, chiefly via the **da Vinci InSite** system (surgeon-controlled, 0° or 30°).

Instrumentation for Grasping and Blunt Dissection

- Mostly **5 mm** (range 3–12 mm). **Single-action** (one jaw moves) versus **double-action** (both jaws move).
- **Tips:** blunt, pointed (dolphin), straight (duckbill), curved (Maryland), angled. **Atraumatic** (serrated or smooth, e.g., 3-cm bowel forceps) versus **traumatic** (toothed). Handles are locking or non-locking; some articulate.
- The suction tip, the heel of the hook and **Kittner (“peanut”) dissectors** (5 and 10 mm) give rapid blunt dissection. **Kittners do not conduct current**, so they can safely lift a line of tissue for division.

Instrumentation for Incising and Hemostasis

- **Cutting mechanisms differ.** **Electrocautery** and **laser** heat cells until internal gas pressure ruptures them. **Ultrasonic devices** use a blade vibrating at **25–55 kHz** over up to 100 µm. **Haemostasis** is always by **denatured protein**: via current or light (heat), or via mechanical vibration (ultrasonic).
- **Scissors:** shorter blades; hooked (sutures), micro (spatulating the ureter), curved (dissection); replaceable tips keep them sharp; some articulate to 90°.

ENERGY DEVICES COMPARED

Device	Mechanism	Lateral thermal spread	Strengths	Hazards
Monopolar (needle, spatula, hook)	Current from the active tip through the patient to a return pad; thinner tip = higher current density	5–7 mm	Versatile; the hook lifts tissue off delicate structures before cutting	Capacitive coupling, insulation failure, direct coupling. Activate only when the whole metal tip is visible and in contact. Active electrode monitoring shuts off on insulation breach
Bipolar	Current only between the jaws	1–6 mm	Less energy; no capacitive coupling	Direct firing only
LigaSure (and EnSeal G2)	Bipolar; impedance feedback ; high current, low voltage → protein seal; built-in blade	1–3 mm	Seals vessels ≤7 mm above physiological pressure; partial nephrectomy margins still readable	One application only (repeats weaken the seal)
Caiman 5	Advanced bipolar, pivoting jaw, electrodes in both jaws	—	Best overall of five sealers in a porcine study	—
Ultrasonic (Harmonic, Sonicbeat, Sonicision)	Piezoelectric crystal → vibration → cavitation, coaptation, coagulation, cutting	≈1 mm	No current through the patient: suits pacemakers and defibrillators and delicate tissue; tissue below 80 °C (less char, fewer adhesions); little smoke; margins readable; cordless versions	Slower sealing; jaws reach >200 °C (bipolar devices stay <100 °C) and take about twice as long to cool (up to 45 s) , so they must not touch bowel
Lasers (CO ₂ , potassium titanyl phosphate (KTP) 400/600 μm, neodymium:yttrium-aluminium-garnet (Nd:YAG), holmium)	Light → heat	—	CO ₂ is rigid (surface vaporisation); KTP and Nd:YAG flexible (non-contact); holmium contact cutting	Largely supplanted in urological laparoscopy; CO ₂ used mainly for endometriosis

- **What the table tests:** thermal spread falls from monopolar (5–7 mm) to bipolar (1–6) to vessel sealer (1–3) to ultrasonic (≈1 mm). The trap: **low spread is not low temperature**. The ultrasonic blade is the hottest instrument after activation and burns bowel on contact.

Instrumentation for Suturing and Tissue Anastomosis

- **Needle drivers:** one fixed jaw; ratchet, spring or Castroviejo lock; straight or pistol grip; **articulating** drivers improve angles. The **Endo Stitch** (10 mm) is used less as skills and robots spread.
- **Lapra-Ty clips:** **3.5-mm absorbable polydioxanone** clips that act as a knot, holding up to **14 days** under low–mid tension. They hold best on **1-0 to 3-0 polyglactin**; **avoid with 4-0** and with tension on 3-0 monofilament (slippage). Uses: single or running sutures, **renorrhaphy bolsters**.
- **Barbed suture** (V-Loc; Quill, which is bidirectional): knotless, does not slide back. Popularised for the **vesicourethral anastomosis**: meta-analysis shows **shorter operative time with comparable complications**.

Instrumentation for Stapling and Clipping

- **Endo GIA** and **ETS-Flex 45** need a **12-mm port** and fire **two triple-staggered rows**, **cutting between them** (ETS: maximum eight loads). Loads are 30, 45 or 60 mm; articulating and roticulating versions exist.

STAPLE CARTRIDGES BY TISSUE

Staple height	Colour	Use
2.0 mm	Grey	Vascular (renal artery and vein)
2.5 mm	White	
3.8 mm	Blue	Thicker tissue (ureter, bowel, bladder)
4.8 mm	Green	

- **Endo-TA (linear, non-cutting)** on the patient side of the donor renal vein gives a **longer donor vein** with no staples to trim. It also closes enterotomies.
- Keep tissue **proximal to the cartridge markers**. **Never fire across clips**. Food and Drug Administration (FDA) reports over 9 years recorded many stapler deaths, injuries and malfunctions.
- **Powered staplers** (from 2010; iDrive Ultra, Echelon Flex GST): more stable and precise, with less bleeding and better economics in bariatric data. Randomised trials are awaited.
- **Clips**: 6–11 mm, through 5–12-mm ports. **No diathermy near clips** (conductive necrosis → clip slip). Closed ends should **project beyond the vessel**, perpendicular to it.
- **Hem-o-lok** (polymer, locking; M, ML, L, XL): up to 10 mm of tissue through a 5-mm port and 16 mm through 10 mm. After transplant-surgeon reports of **renal artery stump haemorrhage**, the manufacturer **contraindicated it for the donor renal artery**. A multi-institutional donor series then found **no bleeding with correct technique** (Box 18.2). A dedicated removal tool exists.

BOX 18.2 BASIC PRINCIPLES OF HEM-O-LOK CLIP PLACEMENT

- **Circumferential dissection** of the vessel → **curved tip seen around and beyond the vessel** (often between artery and vein) → feel the **snap** → **no cross-clipping** → **do not squeeze too hard** → withdraw the sharp applier carefully (it can lacerate the renal vein) → **partial division first** to confirm haemostasis → **at least two clips on the patient side**, leaving a **2-mm arterial cuff** beyond the clips.

From Ponsky L, Cherullo E, Moinzadeh A, et al., Urology 2008.

Instrumentation for Specimen Entrapment and Morcellation

- **LapSac** (1990; nylon with polyurethane lining): **most tear- and leak-resistant**, holds up to **2 kg**, suited to **morcellation**, but difficult to deploy. **Endocatch** (10 and 15 mm): self-opening metal frame, drawstring. **Spring-wire sacks** make scooping easy for **intact removal** through a hand-port or a port enlarged to **5–7 cm**.
- **Manual morcellation**: finger, ring forceps or Kelly clamp. **Ring forceps are preferred** because they are least likely to puncture the sack.

Robotic Instrumentation

- **EndoWrist**: **7 degrees of freedom, 90° articulation**. Most 8-mm instruments have a **10-use limit** (record the “lives” left).
- **Incising and haemostasis**: round-tip and Potts scissors; monopolar **HotShears**, hook and spatula; bipolar Maryland, fenestrated, curved, micro, long and force; **Harmonic ACE**; Vessel Sealer Extend and SynchroSeal (E-100 generator).
- **Graspers**: ProGrasp (**most used in urology**), Cadière, tenaculum, Cobra and others. An **Allen bolt opens locked jaws** manually after power or arm failure.

- **5-mm line:** more limited, and the **5-mm scope is 2D**.
- Needle drivers: standard or **SutureCut** (cuts in the crotch). Clip applicators (titanium, Hem-o-lok ML and L), suction-irrigator, probe stabiliser, retractors.
- **Robotic staplers:** EndoWrist stapler and **SureForm**, which senses **tissue compression and adjusts firing**.
- **Robot failure is rare** (<1%). Half is found before the patient enters, and intraoperative failures (about 0.4%) **rarely force open conversion**.

Exiting the Abdomen

Port Removal and Fascial Closure

- A haphazard exit causes **herniation, incarceration and haemorrhage**.

Exit sequence: lower the pressure to 5 mm Hg → inspect the operative site and every port entry → perfect haemostasis → place fascial sutures under vision → **remove each port under vision** → let CO₂ escape passively through 5-mm sites. **A hand-port is removed and closed first**, then reinsufflate and close the other ports.

FASCIAL CLOSURE RULES

Trocar	Adult	Child
Bladed	Close all 10–12-mm sites (0 polyglactin)	Close all ≥5-mm sites
Non-bladed	Close midline ports >10 mm or stretched sites; some close none at 12 mm if the defect is small	Close 5-mm sites with a single absorbable suture
5-mm ports	Not closed	Closed

- Switching from bladed to blunt trocars cut **abdominal-wall bleeding from 0.83% to 0.16%** and **port-site hernia from 1.83% to 0.19%**.

Instrumentation for Port Site Closure

- Direct closure is hard with BMI >30, so use devices for **en bloc fascia-muscle-peritoneum closure under vision**.
- **Carter-Thomason:** a cone (5, 10 or 12 mm) with two opposed angled channels; a needle-point grasper carries 0 polyglactin in through one channel and retrieves it through the other. It can also act as a “fifth port” (holding a sack, passing a vessel loop).
- **Endo Close:** a spring-loaded suture carrier passed either side of the port.
- **Angiocatheter technique** (cheap): a 14-G sheath passes 0 polyglactin on one side and a **folded 30-inch 0 polypropylene loop** on the other. The loop snares the polyglactin, which is pulled out and tied.

Closure of the Skin

- **Subcuticular 4-0 absorbable** for 10-mm sites. **Adhesive strips** close incisions <10 mm and reinforce those ≥10 mm. **Octyl cyanoacrylate glue** is faster with equal cosmesis.

KEY POINTS: PERFORMING THE PROCEDURE

- Complete the **equipment checklist just before insufflation**; on the robot, plug components into **different circuits** and check the cables.
- **Never insufflate** until all signs of peritoneal entry are confirmed (negative aspiration, easy saline injection, negative saline re-aspiration, positive drop test, normal advancement test).
- **Open access when extensive adhesions are expected**.

- **Non-cutting dilating trocars have superseded bladed ones:** less vessel injury, less herniation.

Concept — The procedure is a sequence of **verified transitions**: every step, from checklist and needle tests to vision-guided trocars and inspection at 5 mm Hg, confirms the previous one before the next begins. Most catastrophic complications come from skipping a confirmation.

PHYSIOLOGIC CONSIDERATIONS IN THE ADULT

- Laparoscopy has been part of operative urology since **Clayman's first laparoscopic nephrectomy**, about 30 years ago. Its physiology centres on the **pneumoperitoneum and positioning**.

Choice of Insufflant

INSUFFLANTS COMPARED

Gas	Advantages	Disadvantages	Use
Carbon dioxide	Colourless, non-combustible , highly blood-soluble (low embolus risk), cheap; absorbed quickly (no lingering distension)	Absorption → hypercarbia, acidosis, arrhythmias ; sympathetic stimulation (↑ heart rate, contractility, resistance); stored in tissues, cleared over hours. COPD patients cannot blow it off	Standard
Nitrous oxide	Less peritoneal irritation; fewer acid-base and arrhythmic effects	Can ↓ cardiac output, ↑ mean arterial pressure (MAP), heart rate and central venous pressure (CVP); supports combustion	Only without electrosurgery
Helium	Inert, non-combustible; no hypercarbia ; less pain and peritoneal irritation; possibly less tumour growth and inflammation	Low blood solubility → gas embolism risk ; expensive; needs a helium yoke	Pulmonary disease; switch to it to salvage intraoperative hypercarbia . Start with CO ₂ , then change
Argon	Inert; argon beam coagulator gas can maintain the pneumoperitoneum	As helium (embolism)	Alternative when hypercarbia occurs
Room air, oxygen	—	Air embolus, combustion, explosion	Abandoned
Xenon, krypton	Inert, non-flammable	Costly; poorly soluble	Not adopted

- **Hypercarbia peaks after extubation**, in recovery, as stored CO₂ is mobilised. Watch patients with lung disease closely after long cases.

Choice of Pneumoperitoneum Pressure

- **15 mm Hg is the commonest setting**, but **12 mm Hg is more physiological** (no change in stroke volume) and **lowers postoperative pain**. **10 mm Hg markedly reduces oliguria**. The authors use **8 mm Hg** for renal surgery with AirSeal.
- **20 mm Hg** raises filling volume by about a fifth and may reduce venous bleeding, but girth barely changes.

TABLE 18.1 PRESSURE EFFECTS: 5, 10, 20, AND 40 MM HG

Effect (supine)	5 mm Hg	10 mm Hg	20 mm Hg	40 mm Hg
Cardiovascular				

Effect (supine)	5 mm Hg	10 mm Hg	20 mm Hg	40 mm Hg
Heart rate	↑	↑	↑	↓
Mean arterial pressure	↑	↑	↑	↑
Systemic vascular resistance	↑	↑	↑	↑
Venous return	↔ / ↓	↓ ↑	↓ ↑	↓
Cardiac output	↔ / ↓	↔ / ↑	↔ / ↓	↓
Renal				
Glomerular filtration rate	↔	↓	↓↓	↓↓
Urine output	↔	↓	↓↓	↓↓
Respiratory				
End-tidal CO ₂	↔	↔ / ↑	↔ / ↑	↑
Arterial partial pressure of CO ₂ (Pco ₂)	↔	↑	↑	↑
Arterial pH	↔	↔ / ↓	↓	↓

- **What it shows:** at 5 mm Hg almost nothing changes. **Renal function is the first casualty**, falling from 10 mm Hg. Afterload (MAP, systemic vascular resistance (SVR)) rises at every pressure. **Venous return and output are preserved or variable up to 20 mm Hg but collapse at 40**, where even heart rate falls. Acidosis builds from 20.
- The body text recommends **≤10 mm Hg to avoid oliguria**, yet the table already shows ↓ GFR and urine at 10. The two are consistent: **10 mm Hg limits, but does not abolish, the renal effect**, which becomes marked (↓↓) at 20.
- **The trap:** patient health and position can override or reverse these supine patterns.

Cardiovascular Effects of the Pneumoperitoneum

Low atrial pressure (normovolaemic or hypovolaemic)	High atrial pressure (hypervolaemic)
Pneumoperitoneum ≤20 mm Hg compresses the cava → venous return falls	The cava resists compression → venous return rises
≥40 mm Hg in either: capacitance vessels collapse, resistance rises, venous return falls sharply . Leg venous return also falls (theoretical deep vein thrombosis (DVT) risk, rare in practice)	

- **Never exceed 20 mm Hg for long; work at 10–12 mm Hg.**
- **Arrhythmias:** hypercapnia → **tachycardia and ventricular ectopics**; peritoneal stretch → **vagal bradyarrhythmia**. Arrhythmia is also an **early warning of pneumothorax, hypoxia or gas embolism**.
- **CVP is unreliable:** intrathoracic pressure inflates readings, so **never base decisions on it**.

Respiratory Effects of the Pneumoperitoneum

- **Pressure-mediated:** the diaphragm is splinted; **dead space is unchanged, functional residual capacity falls**, and **peak airway pressure rises with intra-abdominal pressure**. Use **positive end-expiratory pressure** in lung disease.
- **Head-down position:** diaphragm rises, vital capacity falls. The endotracheal tube may slip into the **right main bronchus**. **Pulmonary oedema** occurs if left heart pressures are high. **Restrict fluids** in long head-down cases (limits facial swelling).

Renal Effects of the Pneumoperitoneum

- **Oliguria and anuria** come from **renal vein compression and direct parenchymal compression, not hormones or ureteric compression**. Cortical flow falls with pressure; medullary flow rises up to 20 mm Hg, then falls.
- **Management:** keep pressure ≤ 10 mm Hg. If needed, give **furosemide, mannitol 12.5–25 g and dopamine 2 $\mu\text{g/kg/min}$** to keep urine output **>100 mL/hour**, rather than fluid boluses, which cause overload and oedema.

Effects on Mesenteric Blood Flow and Intestinal Motility

- Splanchnic flow falls (mesenteric vessels, liver, pancreas, stomach, spleen, bowel). Rarely, **mesenteric thrombosis** develops over days.
- Laparoscopy causes **less ileus than open surgery**, perhaps via hypercarbia. Gut perfusion is unchanged at 15 mm Hg. Bacterial translocation rises with pressure in rats. There is **no increase in reflux or regurgitation**.

Acid–Base Metabolic Effects of Pneumoperitoneum

- Respiratory acidosis comes from **transperitoneal CO_2 absorption** (dead space is unchanged). In normal patients, increase minute ventilation; in COPD it is dangerous.
- **COPD: arterial blood gases for any procedure with >1 hour of insufflation, continued in recovery.**
- **CO_2 is absorbed transperitoneally** through the peritoneum, and **extraperitoneally** through fat, connective tissue and disrupted lymphatics. Absorption **rises over the first 30–60 minutes, then plateaus**.
- Which route absorbs more is debated. A **sealed balloon-tip cannula removes any clinical difference**.
- **Monitor end-tidal CO_2 and saturation**; check gases in long cases and high-risk patients (airway disease, renal failure, heart failure, old age).
- **Rising end-tidal $\text{CO}_2 \rightarrow$** the anaesthetist raises rate and volume, and the surgeon **lowers the pressure or desufflates**.

Hemodynamic Effects Related to Patient Position and Type of Approach

HAEMODYNAMICS BY POSITION

Position	Heart rate	MAP	SVR	Cardiac output
Supine, <15 mm Hg	—	$\uparrow$	$\uparrow$	$\leftrightarrow$ or $\downarrow$
Supine, >20 mm Hg	—	$\downarrow$	—	$\downarrow$ (venous return falls)
Head-up	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$
Head-down (including steep Trendelenburg)	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$

- **Head-down is haemodynamically favourable** (higher output via venous return), despite its respiratory cost.
- **Retroperitoneal versus transperitoneal:** transperitoneal causes larger swings in cardiac output, pulmonary artery pressure, CVP and iliac and caval pressures. In the pelvis, **extraperitoneal insufflation produced higher arterial CO_2** and needed more minute ventilation.

Hormonal and Metabolic Effects and Immunologic Effects

- **Stress hormones rise** as in any surgery (β -endorphin, cortisol, prolactin, catecholamines). Vasopressin rises with insufflation (significance unknown).

- Compared with open surgery (cholecystectomy data): **less hyperglycaemia, less insulin resistance, a smaller hepatic stress response and less nitrogen catabolism**, which may underlie faster convalescence.
- **Less immunosuppression** (C-reactive protein, interleukin-6, T-cell subsets, natural killer cells, skin tests), though no difference was found for laparoscopic versus open nephrectomy for cancer. A long-term oncological benefit is unproven.

KEY POINTS: PHYSIOLOGIC CONSIDERATIONS IN THE ADULT

- **CO₂ is standard:** non-combustible and rapidly absorbed.
- **Helium suits lung disease**, where hypercarbia is poorly tolerated.
- **Do not exceed 20 mm Hg for long; work at 10–12 mm Hg.**

Concept — Pneumoperitoneum physiology has two drivers: **pressure** (caval, renal and diaphragmatic compression) and **absorbed CO₂** (acidosis, sympathetic drive). Every recommendation in this section lowers one or the other, and every high-risk patient (COPD, hypovolaemia, renal impairment) is one who tolerates one of them badly.

COMPLICATIONS AND TROUBLESHOOTING IN LAPAROSCOPIC AND ROBOTIC SURGERY

- Historically, about **4% complications** and **mortality 0.03–0.08%**. Contemporary series show **13–22%**, reflecting more complex procedures. **Vascular, then adjacent-organ injuries are commonest.**

TABLE 18.2 MAJOR COMPLICATIONS OF TRANSPERITONEAL ABDOMINAL SURGERY

Outcome	Abdominal series (Parsons, 2004)	Mainly pelvic series (Vallancien, 2002)
Overall complications	13.2%	22.6%
Intraoperative / postoperative	5.7% / 7.5%	3.6% / 19%
Death	0.2%	0%
Vascular injury	2.8%	0.5%
Bowel injury	1.1%	1.2%
Adjacent organ injury	1.1%	0.8%
Conversion	1.7%	1.7%

Data from Parsons JK, et al., Urology 2004 (894 procedures, all abdominal); Vallancien G, et al., J Urol 2002 (1,311 procedures, 84% pelvic).

- **Pattern: upper-abdominal (retroperitoneal-organ) surgery carries the vascular risk**, about five times the pelvic rate. **Pelvic surgery accumulates postoperative complications.** Bowel injury and conversion are similar. The trap: the higher pelvic total is **mostly postoperative**, not intraoperative.

Minimizing the Incidence of Complications During the Learning Curve

- Start with **low-risk patients of normal habitus.**
- **Three-arena training** (Society of American Gastrointestinal and Endoscopic Surgeons): **(1) courses** (didactic, live cases, hands-on laboratory); **(2) preceptorship**, watching **five or more cases**; **(3) mentoring** of the first cases.
- Add **box trainers** (proven benefit for cutting, clipping and suturing), **1-week mini-residencies** (most graduates go on to complex laparoscopy), and **virtual reality (VR)**, which beats no training or box training for novices.

General Procedural Complications

Malfunction of Equipment

- Trained staff must recognise and fix faults. Keep the Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) troubleshooting guide and **vendor contacts** available, and train staff in-depth on integrated and robotic systems.
- **da Vinci malfunction about 0.4%:** most found before the case (cancelled); few converted to laparoscopy or open surgery.

Complications Related to Obtaining the Pneumoperitoneum

VERESS AND OPEN-ACCESS COMPLICATIONS

Complication	Recognition	Management	Prevention
Preperitoneal insufflation	Steep pressure rise at about 500 mL , asymmetric distension. If missed, 1–2 L can mimic success; the camera shows only fat	Evacuate via the side arm → open (Hasson) entry	Initial pressure above the 10 mm Hg limit = preperitoneal; advancement test (0.5–1 cm free)
Vascular puncture	Blood in the hub or on aspiration	Withdraw without manipulation ; re-site or go open. Inspect the track at 5 mm Hg ; pressure and haemostatic agent. Unexplained instability or loss of space = hidden retroperitoneal haematoma	Aim towards the pelvis at the umbilicus; 12-mm incision, spread fat, steady fascia with Allis clamps (vital in children); non-umbilical sites
Visceral puncture	Aspirate blood, urine or bowel content; high pressure in a solid organ	Withdraw and re-site. Liver or spleen: pressure, argon beam, haemostatic agent (general surgery if needed). Bowel or bladder needle puncture: withdrawal only	Nasogastric tube and bladder catheter first
Open-access injury	Bowel adherent under the wall	Repair through the same incision if seen early	Less frequent than closed; vascular injury still possible

Complications Related to Insufflation and Pneumoperitoneum

INSUFFLATION COMPLICATIONS

Complication	Recognition	Management	Prevention
Bowel insufflation	Asymmetric distension, flatus , high pressure after <2 L	Disconnect (gas outflow confirms) → withdraw → open access elsewhere	Needle tests; avoid scarred sites; open access
Gas embolism	Sudden cardiovascular collapse ; arrhythmia, cyanosis, pulmonary oedema. End-tidal CO₂ abruptly rises, saturation falls, then end-tidal CO₂ falls sharply . “Millwheel” murmur; foamy blood	Stop insufflation, desufflate → left lateral decubitus (right side up) , head down (gas trapped in the right ventricular apex) → 100% oxygen, hyperventilate → aspirate via a central line; hyperbaric oxygen or bypass reported	Commonest cause: insufflating through a needle in a vessel or organ. Never insufflate with the slightest doubt
Barotrauma (>15 mm Hg prolonged)	Hypotension (caval compression); pneumothorax or pneumomediastinum ; rising ventilation pressure; may worsen a hiatal hernia	Desufflate → restart at 10 mm Hg ; replace a faulty insufflator; open a side arm with the argon beam or CO₂-cooled laser	Check the insufflator before every case; monitor pressure. PEEP can rupture blebs

Complication	Recognition	Management	Prevention
Subcutaneous emphysema	Crepitus over abdomen and chest; pneumoscrotum	Misplaced needle → open. Port leak → purse-string, larger or balloon trocar , lower pressure	Incisions tailored to the port; ports aimed at the target. Commoner retroperitoneally → use 12 not 15 mm Hg and a balloon trocar
Pneumopericardium (about 0.8%)	Impaired cardiac function; usually found on the recovery chest film	As for gas embolism; pericardiocentesis for tamponade	Pressure 12 mm Hg ; tight, well-seated ports; all trocars below the 12th rib ; know the diaphragm's relationships
Pneumomediastinum	Usually silent	—	
Pneumothorax (1.6–4%; commoner retroperitoneally)	Neck and chest emphysema; tension : hypotension, absent breath sounds, high ventilatory pressure	16-G needle, second or third intercostal space, midclavicular line → chest drain	

Complications Related to Initial “Blind” Placement of the First Trocar

- **Non-bladed and optical trocars** have markedly reduced catastrophic injuries.
- **Gastrointestinal: small or large bowel is commonest**; liver and spleen are rarely injured (lateral position).

One-wall bowel injury	Through-and-through bowel injury
Mucosal folds seen as soon as the scope enters	Seen only when the scope is passed through a secondary port to view the primary site, which must be done routinely . If missed until exit → wider injury, contamination, peritonitis or death
Leave the trocar in place → open access elsewhere → repair (open or laparoscopic)	Repair similarly

Both: general surgery consultation; irrigate with 4–5 L antibiotic saline; broad-spectrum antibiotics. Stomach: rare; primary closure; decompress with a nasogastric tube

- **Major vascular injury** (0.11–2%): **commoner in retroperitoneal (upper-tract) than pelvic laparoscopy**. The **aorta and common iliac arteries** are most often injured; the **cava** less (lateral to the aorta) and the **common iliac vein** rarely (posterior to the artery). Also mesenteric vessels of fixed loops and epigastric vessels.
- **Signs**: sudden **hypotension and tachycardia**; pulsatile or venous blood from the sheath; or rapid haemoperitoneum, mesenteric haematoma or a **shrinking space from retroperitoneal haematoma**.

Major vessel injury: call a **vascular or trauma surgeon** → **close the trocar and leave it in** → **emergency laparotomy following the trocar** to the vessel → proximal and distal control (vessel loops, bulldogs) or **Satinsky** → withdraw the trocar and repair. If the trocar is already out: **swing the sheath and scope to the abdominal wall and cut down on it**, or convert to **hand-assist** and compress.

- **Prevention**: know the vascular anatomy and **review the CT** for anomalies; extra care in children and thin adults. Confirm Veress signs; achieve adequate pneumoperitoneum (**20 mm Hg temporarily** for the first trocar); **optical, blunt trocars**; **avoid scars**.

BOX 18.3 CONTENTS OF HEMORRHAGE TRAY FOR LAPAROSCOPIC SURGERY

- **Laparoscopic Satinsky clamp** · **10-mm suction-irrigation tip** (clot) · **Endo Stitch** with 4-0 absorbable suture · **Lapra-Ty** applier and clips · **6-inch 4-0 vascular suture** on a small half-circle (SH) needle with a

Lapra-Ty preplaced · two needle drivers · topical haemostatic agent. With it, some major venous injuries can be repaired laparoscopically.

- **Urinary tract injury:** mostly **bladder injury at first trocar placement** (wide range in gynaecological series; reduced by blunt trocars). It presents with **pneumaturia or haematuria**. Confirm with **intravesical indigo carmine**; **always repair**, laparoscopically or open, **never catheter drainage alone**. **Prevention: urethral catheter in every major case.**

Complications Related to Placement of Secondary Trocars

- **Port-site bleeding:** blood drips onto the viscera. Find it by **cantilevering the trocar into each quadrant** until it tamponades.
- **Fixes:** **diathermy** via another port (costliest); a **straight Keith needle with 0 absorbable suture** passed in and out across the quadrant and **tied over a gauze bolster** (cheapest; the port stays usable); or a **Carter-Thomason suture**. Finally close the site with a closure device.
- **Prevention:** **transilluminate**; spread subcutaneous tissue with a Kelly clamp; **blunt trocars** (epigastric injury about fivefold less; bleeding 3% vs 9%); place ports **in the midline or ≥ 6 cm lateral to it**.

TROCAR POSITION PROBLEMS

Problem	Cause	Remedy
Crossing swords	Ports too close: intra-abdominal shafts cross	Move the handles together (tips move apart)
Striking handles	Ports too close: external handles collide (robotic equivalent: arm collision ; keep ports 8–10 cm apart)	Withdraw sheaths slightly
Rollover	Scope and instrument ports point at each other , so the instrument deflects off the scope	Move the handles together; 30° lens parallel and rotated

- These are worst in **laparoendoscopic single-site surgery (LESS)**, which needs articulating instruments, differing lengths and a low-profile scope. Prevent them with planning: **in-line ports for pyeloplasty**, a **diamond for nephrectomy**, each port pointed at the pathology. Add a 5-mm port without hesitation.

Complications Related to the Surgical Procedure

FOUR MECHANISMS OF ELECTROSURGICAL BOWEL INJURY

Mechanism	How it happens	Prevention
Inappropriate direct activation	Instrument left unobserved or out of view; activated by someone other than the surgeon; thermal spread at high settings (>30 W)	Remove instruments when idle; only the primary surgeon activates ; isolate the target; prefer bipolar
Direct coupling	Active electrode touches another metal instrument that contacts bowel, often out of view	Activate only with the whole metal portion visible , including exposed joints behind the tip
Capacitive coupling	Charge induced around the electrode cannot disperse: metal cannula in a plastic retainer , or an electrode through a metal operating laparoscope in a plastic sheath. Worse with old high-voltage generators, thick electrodes, coagulation mode	Never mix conducting and non-conducting elements ; modern generators, thin electrodes, blend or cut current
Insulation failure	Breaks from reuse, resterilisation or trocar trauma give high current density at the defect	Inspect insulation before use; active electrode monitoring (instant shutdown)

- **Recognition:** whitish serosal spots, but often **missed**. **Necrosis can take up to 18 days**, so symptoms may start after discharge.

- **Investigate any fever after day 1 or increasing pain.** Atypical signs: low-grade fever, **leukopenia**, **severe pain at the port nearest the injury**. A left shift is more reliable than the white count.
- Plain films mislead: **free CO₂ persists up to 9 days**. The test is **CT with oral contrast and delayed films**.

BOWEL INJURY: MANAGEMENT BY MECHANISM

Injury	Presentation	Management
Monopolar thermal	Late (days to 18 days)	Damage extends beyond the visible necrosis → resect with a 6-cm margin each side and anastomose end to end. A minor late injury (>5–7 days) may be managed with antibiotics and elemental diet (closed fistula), but explore if not improving
Bipolar thermal	Direct firing only	Confined to the visible area : small → excise and close; >half the circumference → segmental resection
Mechanical (graspers, scissors)	Usually seen intraoperatively ; if missed, early fever , ileus, peritonitis	Laparoscopic repair (no bowel preparation needed; resection rarely needed) plus 4–5 L antibiotic irrigation. Missed → immediate re-operation , excise or resect

- **Mechanical prevention**: atraumatic graspers on bowel; instruments in under vision, out when idle.
- **Vascular injury during dissection** is rare (blunt trocars, small instruments, magnification). The **right renal dissection** puts the **cava, renal and gonadal veins** at risk.

Venous injury: raise pressure to 25 mm Hg → clear with the suction-irrigator and identify the site → **sponge or pledget tamponade** → haemostatic sealant (fibrin glue, gelatin-thrombin) → **extra 5-mm port** for two-handed suturing or Satinsky → or **hand-assist**: **pinch the vein's side walls closed** (pressing on top enlarges the hole), slide a Satinsky under the fingers → or convert, **tamponading with forceps while cutting down on a swung port**. **Keep the patient well filled**: hypovolaemia at high pressure risks gas embolism.

- **Arterial injury**: minor → tamponade. **Renal artery during nephrectomy** → **staple it**. **Aortic** → **convert**.
- **Nerve injury (about 2.8% in a multicentre survey)** comes from position plus duration. The **brachial plexus is most at risk**: abduction >90°, external rotation of the humerus, and **shoulder braces in Trendelenburg** (clavicle driven back), especially in steep-Trendelenburg robotic prostatectomy. Also the **femoral nerve** (lithotomy hip abduction and rotation) and the **sciatic nerve** (upper leg in lateral).
- Nerve injury appears on **day 1 when mobilising**. Obtain an **early neurology consultation** and nerve conduction studies (documentation), then physiotherapy. If not better in days, recovery takes months.
- **Prevention**: table accessories; arms at the side **pronated**; pad all lateral pressure points with a pillow between the legs; pad under straps. **No shoulder braces**: use **wide padded chest straps or a beanbag**. Avoid extreme hip abduction; **recheck padding after every table move**.

Complications Related to Exiting the Abdomen

- **Bowel entrapment** at a port: **ileus and point tenderness on day 2–3**. Treat by re-insufflating via another site, three ports, atraumatic clamps either side, opening the site and **reducing by fingertip**. Resection is rarely needed.
- Entrapment is a **technical error**. Ports have a **side hole near the tip that equalises pressure** on withdrawal. **Inspect each site on removal**, and **back the last port off over the scope so the scope leaves last**.
- **Late-declared port bleeding**, unmasked when the trocar's tamponade is removed → inspect every site at 5 mm Hg.

Early Postoperative Complications

- **Pain:**
 - **Port-site pain** is normal early; if increasing → **hernia**. A port-site bulge → rectus sheath haematoma, port haematoma or hernia. Pain without swelling → broad fascial suture or a palpable knot.
 - **Immediate severe diffuse pain** → noxious spill (e.g., polycystic kidney fluid) or **bowel injury**.
 - **Shoulder-tip pain, right posterior shoulder** = diaphragmatic CO₂ irritation, occasionally mimicking pulmonary embolism.
 - **Delayed diffuse pain with low-grade fever** → occult bowel injury.
- **Incisional hernia:** in adults, **ports >10 mm**; in children, **even 5 mm**. Localised pain, nausea, ileus; diagnose by CT (bowel above the fascia). Repair laparoscopically, with general surgery if strangulated.
- **Hernia prevention:** blunt trocars; closure rules as tabulated above. **Hand-assist midline sites herniate in 4–7%:** close with **interrupted non-absorbable sutures**. **Transverse beats vertical** midline fascial incisions; in robotic prostatectomy, switching to horizontal camera and extraction incisions cut hernias **from 5.4% to 0.4%**.
- **DVT and pulmonary embolism (PE): no more common than after open surgery**, and no randomised trials exist.
 - **American Urological Association (AUA) Best Practice Statement: pneumatic compression for all**, from before surgery to **48–72 hours** after. **High risk** (morbid obesity, smoking, previous DVT): add **heparin 5,000 units 2 hours before and every 12 hours after**.
 - In upper retroperitoneal surgery, compression alone performed as well as heparin, and heparin may add bleeding (non-randomised data).
- **Wound infection:** rare, except at **hand-assist sites (about 9%)**. Prevent with antisepsis, port irrigation and careful closure.
- **Rhabdomyolysis** (about 0.4% overall, **about 1% retroperitoneal**): **men, long renal cases, kidney rest throughout**. It presents **in recovery** with **severe downside hip pain and brown urine**, with **creatinine kinase >5,000 units**. Padding is unproven. **Prevent by limiting the kidney rest to <1 hour, avoiding hypotension, and keeping cases under 5 hours**.

Late Postoperative Complications

- After 3 weeks, complications are rare: mainly **lymphatic** and **incisional hernia**.

KEY POINTS: COMPLICATIONS AND TROUBLESHOOTING IN LAPAROSCOPIC AND ROBOTIC SURGERY

- **Start with low-risk, normal-habitus patients.**
- **Gas embolism:** abrupt end-tidal CO₂ rise, saturation fall, then end-tidal CO₂ collapse.
- **Four thermal mechanisms:** direct activation, direct coupling, capacitive coupling, insulation failure.
- **Plan ports** to avoid crossing swords, striking handles, rollover and arm collisions.

Concept — Complications follow the operation's chronology: **entry** (needle, first trocar), **insufflation** (gas in the wrong place), **dissection** (energy and traction), **exit** (entrapment, bleeding) and **recovery** (hernia, rhabdomyolysis, delayed bowel necrosis). Each phase has a characteristic sign, and **the dangerous injuries are the ones that present days later**.

TRAINING AND PRACTICING LAPAROSCOPIC AND ROBOTIC SURGERY

Equipment for Practicing Laparoscopic and Robotic Surgery

- Models range from inanimate boxes to VR, live animals and cadavers. **Use non-living models unless a live one is essential.**
- **Training box:** opaque top with ports, monitor, light source, camera. Commercial boxes are costly; **mobile-phone and 3D-printed options** (e.g., iTrainer) exist.
- Set up as in theatre: **camera between the instruments, 25–45° between instruments, <55° to the horizontal.** A **camera driver** also trains camera skills. Practise with **3-0 suture on an SH needle.**
- **Inanimate drills** use paper, sponge, glove and tubing. **Non-living animal tissue** is realistic and cheap but messy. A **chicken oesophagus and stomach** works for vesicourethral anastomosis and pyeloplasty practice.

TABLE 18.3 LIST OF TRAINING MODELS FOR LAPAROSCOPIC AND ROBOTIC SURGERY — REBUILT BY TYPE

Type	Models (origin)	Procedure trained
Bench	Partial nephrectomy model (UC Irvine); latex model (McMaster); hydrogel reimplant model (Cleveland Clinic); hydrogel partial nephrectomy model (Rochester, perfused, 3D-printed)	Laparoscopic partial nephrectomy (LPN); laparoscopic vesicourethral anastomosis; laparoscopic ureteric reimplant (LUR); robotic partial nephrectomy (RPN)
Virtual reality	LapSim (Surgical Science); ProCedicus MIST (Mentice); LapMentor (Simbionix); SimNow (Intuitive) ; RobotiX Mentor (Simbionix)	LPN (laparoscopic simulators); robotic-assisted radical prostatectomy (RARP) (SimNow); basic robotic skills

- **Pattern: bench models train specific reconstructive steps** (anastomosis, renorrhaphy) with real tissue feel, whereas **VR trains and measures skills and whole procedures.** Partial nephrectomy dominates both, because it combines excision, haemostasis and suturing. The trap: **VR still emulates tissue poorly.**
- **Structured courses:**
 - **European Association of Urology (EAU) E-BLUS** (Basics in Laparoscopy, since 2013): online theory → tutored dry-model training (**peg transfer, circle cutting, needle guidance, suturing**) → examination and certification. Uptake is rising, but residents' skills remain low.
 - **Fundamentals of Robotic Surgery (FRS)** (Department of Defense and Intuitive): proficiency-based modules (system, console ergonomics, suturing, team communication). A multi-institutional randomised controlled trial showed **better speed and accuracy** after training.

TRAINING MODALITIES COMPARED

Modality	Advantages	Limitations
Box trainer	Cheap; proven skill gains	Limited realism
Virtual reality	Tracks metrics (precision, economy of motion), builds curricula, improves live performance and operating time for novices; expert benchmarks; robotic simulators help maintain skills during inactivity and correlate with live RARP performance	High cost; poor tissue emulation
Live animal	Realistic anatomy; real haemostasis. Swine are commonest (size, availability, similar urinary tract); rabbits for fine work	Veterinary facilities, anaesthesia, euthanasia, cost. Pig kidney bleeds unrealistically little; porcine bladder is almost entirely intraperitoneal. Reserved for courses

Modality	Advantages	Limitations
Cadaver	Most realistic anatomy: no substitute	Cost; donor programme needed

Formal Training Programs

- **Competence = technical plus non-technical skills** (judgement, anticipation, economy of motion). Residency has been more “learning” than “teaching”. Safety, efficiency and cost pressures favour formal programmes.
- **Mini-residency or mini-fellowship** (typically **5 days**): didactics, case observation, inanimate and animate laboratory, then **proctoring**. About **three-quarters** were doing laparoscopic renal surgery at 3 years, and **about 90%** robotic prostatectomy after a 5-day course.

KEY POINTS: TRAINING AND PRACTICING LAPAROSCOPIC AND ROBOTIC SURGERY

- **Skills training outside the operating room** prepares the learner for safe, effective surgery.

Concept — *Training moves skill acquisition **off the patient**: from box to VR to animal or cadaver, then proctored practice. Each step adds realism at greater cost, and validated metrics decide progression.*

CONCLUSION

- Procedures unthinkable in 1990 are now routine. The **robot extends complex reconstruction to more urologists** and often improves precision and outcome.
- Mastery of the basics (pneumoperitoneum physiology, access, complication management) is what makes this possible: **Osler's “treatments that harm less.”**